



Startup Zone

1 Augmented Audio: The Next Big Thing In Cinemas

2 The Do's And Don'ts For A Consumer Electronics Startup

3 Empowering The World With Solar Energy

4 Smallsat Revolution: New Era In Space industry

1 AUGMENTED AUDIO: THE NEXT BIG THING IN CINEMAS

Cinema is one of the most popular forms of entertainment worldwide. We can watch movies not only in theatres but also on devices such as TVs, laptops, cellphones and projectors. What makes the demand for watching movies in theatres so high?

The answer lies in the design of a movie theatre, which provides the audience with a unique cinematic experience. Everything from lighting and sound to seating arrangement is taken care of to provide maximum comfort. Still, many a time, we are unable to hear the audio due to disturbances in surroundings.

To tackle this problem, Finland-based startup Flexound, a member of International Cinema Technology Association (ICTA), integrated its patented technology Flexound Augmented Audio into theatre seating—it is now available globally.

The company was founded in 2015 by acoustics specialist Jukka Linjama, along with Mervi Heinaro and Tommi Immonen. They initially worked to help autistic children by adding the sensation of touch to music, but later switched to the digital entertainment market owing to its potential. Augmented Audio module is scalable and easy to integrate. It can be embedded into cushions, car seating, furniture and many more areas, thereby providing relaxation at even low volume.

The major benefit of this technology is that sound quality remains the same, no matter where you are seated in the theatre, and only one module is required per seat. It also enables lower sound volume, and can be used with any pre-existing sound system without much hassle. You do not need any



(Credit: www.epressi.com)

wearable accessories to get clarity in listening. It requires low maintenance, and has a long lifespan. Energy consumption is low, too.

Launched in 2017, the company's first consumer product, Humu smart cushion is a near-field listening device that contains the technology module with elastic vibrating elements to connect to the audio source of any audio-visual material. This is useful for both entertainment and health (like meditation) purposes. Sound and vibration frequency range from 20Hz to 20kHz.

After initial success, Augmented Audio premiered in CineEurope, Barcelona, in June 2018, and the first-ever commercial movie theatre in Finland began using the technology in February 2019.

After extensive testing at a pilot cinema, it was found that high-quality surround sound combined with physical vibrations enhanced the experience of consumers and they were ready to pay for it. The company then negotiated with exhibitors worldwide.

Finally, Ferco Seating, a manufacturer of public seating across different venues such as cinema and auditoriums, launched two certified models—Premium Verona and Premium Milano—from the startup.

Verona is a fully reclining chair that allows users to adjust the position as per need, and it is available as a single seater, twin seat or link seat

with shared armrests. The optional easy-lift system and under-seat lighting enable effective cleaning. Other features include integrated tables, built-in cup holders and charging points.

Comfortable and aesthetically pleasing, Milano is available as a single or twin seat with an optional rocker-back mechanism that allows the seat to be manually reclined to any angle. Other features include footrest sensor, auto return footrest, built-in cup holders, lighting and USB charging for electric recliners. It is good for theatres where both capacity and comfort need to be kept in mind.

Pricing depends on drawing/layout, product choice, fabric and finishing, among others. Flexound also has its own Finland-produced model.

The company is further planning on more models and manufacturers. This gives operators an edge in their business when compared to in-home entertainment and other theatres, as more people enjoy going to cinema halls.

—Ayushee Sharma